

Statistics & Probability



Permutations

- 1 How many ways can 5 distinct books be arranged on a shelf?
 - 2 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?
 - 3 How many distinct arrangements are there of the letters in the word APPLE?
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Combinations

- 4 How many ways can a committee of 4 be chosen from 11 people?
 - 5 A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?
 - 6 From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?
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Probability rules

- 7 A fair die is rolled. Find $P(\text{even or greater than 4})$.
 - 8 $P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.
 - 9 A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.
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Conditional probability and independence

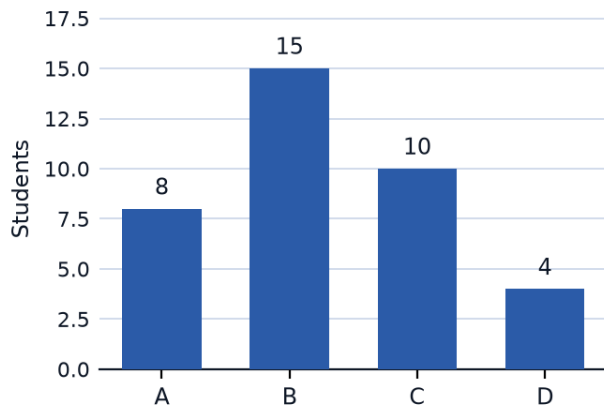
- 10 $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?
 - 11 In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.
 - 12 Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.
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Measures of center and spread

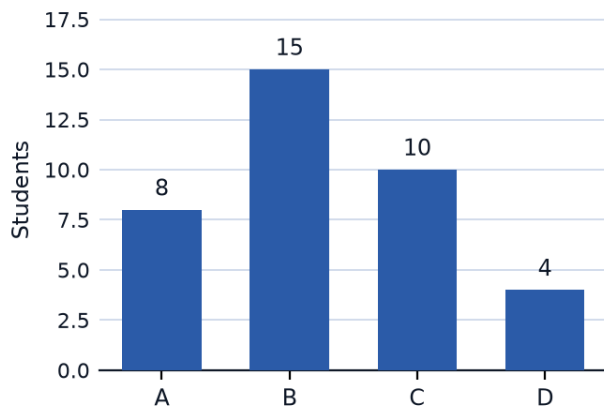
- 13 Find the mean and median of the data set 12, 15, 18, 20, 25.
- 14 Find the population standard deviation of the data set 4, 8, 6, 5, 3.

Data representation

- 15 The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.



- 16 Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.



The binomial theorem

- 17 Find the coefficient of x^3y^2 in the expansion of $(x + y)^5$.
- 18 In the expansion of $(2x - 1)^4$, find the coefficient of x^2 .

The normal distribution

- 19 A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?
- 20 For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.
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A well-designed word problem: quality control sampling

- 21 This question has three parts. A factory produces batteries, and 4% of batteries are defective. A quality-control inspector randomly selects a sample of 3 batteries from a large shipment (assume the selections are independent). (a) Find the probability that none of the 3 batteries are defective. (b) Find the probability that exactly 1 of the 3 batteries is defective. (c) Find the probability that at least 1 of the 3 batteries is defective.
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Multiple choice — easy

- 22 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 8 members?
- a. 512
 - b. 330
 - c. 342
 - d. 336
- 23 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 18 members?
- a. 4902
 - b. 4890
 - c. 4896
 - d. 5832

- 24** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 16 members?
- a. 3360
 - b. 4096
 - c. 3366
 - d. 3354
- 25** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 13 members?
- a. 2197
 - b. 1722
 - c. 1716
 - d. 1710
- 26** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 10 members?
- a. 714
 - b. 726
 - c. 720
 - d. 1000
- 27** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 14 members?
- a. 2178
 - b. 2184
 - c. 2190
 - d. 2744

- 28** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?
- a. 1326
 - b. 1314
 - c. 1320
 - d. 1728
- 29** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 20 members?
- a. 6834
 - b. 6840
 - c. 6846
 - d. 8000
- 30** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 17 members?
- a. 4913
 - b. 4080
 - c. 4086
 - d. 4074
- 31** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 9 members?
- a. 504
 - b. 498
 - c. 729
 - d. 510

- 32** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 19 members?
- a. 6859
 - b. 5808
 - c. 5814
 - d. 5820
- 33** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 7 members?
- a. 343
 - b. 210
 - c. 204
 - d. 216
- 34** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 6 members?
- a. 216
 - b. 114
 - c. 120
 - d. 126
- 35** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 15 members?
- a. 2736
 - b. 3375
 - c. 2730
 - d. 2724

- 36 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 11 members?
- a. 996
 - b. 984
 - c. 1331
 - d. 990
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Multiple choice — medium

- 37 How many ways can a committee of 6 be chosen from 11 people?
- a. 330
 - b. 462
 - c. 465
 - d. 459
- 38 How many ways can a committee of 5 be chosen from 11 people?
- a. 465
 - b. 459
 - c. 462
 - d. 467
- 39 How many ways can a committee of 4 be chosen from 15 people?
- a. 1365
 - b. 1362
 - c. 3003
 - d. 1368

- 40 How many ways can a committee of 5 be chosen from 19 people?
- a. 11631
 - b. 27132
 - c. 11628
 - d. 11625
- 41 How many ways can a committee of 4 be chosen from 14 people?
- a. 1004
 - b. 1001
 - c. 998
 - d. 2002
- 42 How many ways can a committee of 5 be chosen from 7 people?
- a. 21
 - b. 24
 - c. 18
 - d. 7
- 43 How many ways can a committee of 4 be chosen from 17 people?
- a. 2383
 - b. 6188
 - c. 2380
 - d. 2377

- 44 How many ways can a committee of 3 be chosen from 10 people?
- a. 117
 - b. 123
 - c. 210
 - d. 120
- 45 How many ways can a committee of 4 be chosen from 20 people?
- a. 4842
 - b. 4845
 - c. 4848
 - d. 15504
- 46 How many ways can a committee of 3 be chosen from 6 people?
- a. 20
 - b. 23
 - c. 17
 - d. 15
- 47 How many ways can a committee of 5 be chosen from 16 people?
- a. 8008
 - b. 4371
 - c. 4365
 - d. 4368

- 48 How many ways can a committee of 2 be chosen from 18 people?
- a. 156
 - b. 150
 - c. 153
 - d. 816
- 49 How many ways can a committee of 4 be chosen from 7 people?
- a. 21
 - b. 32
 - c. 38
 - d. 35
- 50 How many ways can a committee of 3 be chosen from 19 people?
- a. 972
 - b. 969
 - c. 3876
 - d. 966
- 51 How many ways can a committee of 4 be chosen from 10 people?
- a. 213
 - b. 252
 - c. 210
 - d. 207

Multiple choice — hard

- 52 Events C and D are independent, with $P(C) = 0.2$ and $P(D) = 0.9$. Find $P(C \cup D)$.
- a. 0.97
 - b. 0.82
 - c. 0.92
 - d. 1.02
- 53 Events C and D are independent, with $P(C) = 0.4$ and $P(D) = 0.2$. Find $P(C \cup D)$.
- a. 0.62
 - b. 0.52
 - c. 0.57
 - d. 0.42
- 54 Events C and D are independent, with $P(C) = 0.3$ and $P(D) = 0.3$. Find $P(C \cup D)$.
- a. 0.56
 - b. 0.41
 - c. 0.61
 - d. 0.51
- 55 Events C and D are independent, with $P(C) = 0.2$ and $P(D) = 0.8$. Find $P(C \cup D)$.
- a. 0.74
 - b. 0.84
 - c. 0.89
 - d. 0.94

- 56 Events C and D are independent, with $P(C) = 0.6$ and $P(D) = 0.6$. Find $P(C \cup D)$.
- a. 0.94
 - b. 0.89
 - c. 0.84
 - d. 0.74
- 57 Find the coefficient of x^4y^6 in the expansion of $(x + y)^{10}$.
- a. 210
 - b. 212
 - c. 208
 - d. 120
- 58 Find the coefficient of x^7y^3 in the expansion of $(x + y)^{10}$.
- a. 118
 - b. 122
 - c. 120
 - d. 210
- 59 Find the coefficient of x^1y^5 in the expansion of $(x + y)^6$.
- a. 6
 - b. 1
 - c. 4
 - d. 8

- 60 Find the coefficient of x^6y^4 in the expansion of $(x + y)^{10}$.
- a. 210
 - b. 212
 - c. 252
 - d. 208
- 61 Find the coefficient of x^6y^3 in the expansion of $(x + y)^9$.
- a. 82
 - b. 126
 - c. 86
 - d. 84
- 62 From a group of 10 men and 5 women, how many ways can a committee of 6 be formed with exactly 4 men and 2 women?
- a. 2110
 - b. 2100
 - c. 220
 - d. 2090
- 63 From a group of 10 men and 8 women, how many ways can a committee of 5 be formed with exactly 3 men and 2 women?
- a. 148
 - b. 3350
 - c. 3370
 - d. 3360

- 64 From a group of 10 men and 5 women, how many ways can a committee of 5 be formed with exactly 3 men and 2 women?
- a. 1210
 - b. 130
 - c. 1190
 - d. 1200
- 65 From a group of 7 men and 7 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?
- a. 42
 - b. 441
 - c. 431
 - d. 451
- 66 From a group of 10 men and 7 women, how many ways can a committee of 7 be formed with exactly 4 men and 3 women?
- a. 7340
 - b. 7360
 - c. 7350
 - d. 245